

Hamro Hospital Management System - Complete Project Documentation

Prepared for final production delivery. This document is suitable for technical teams, hospital management, administrators, and deployment engineers.

1. Project Overview

Hamro Hospital Management System is a role-based Django hospital information system built for patient registration, doctor consultation, departmental workflows, billing, reporting, staff attendance, leave, payroll, backup, and production deployment readiness.

- Approved homepage and dashboard design preserved.
- Role-based access control implemented for hospital staff and patients.
- Barcode-driven patient identification is central to patient lookup and printed documents.
- SQLite is retained for local development and PostgreSQL is supported for production.

2. System Architecture

The system follows a modular Django architecture. Each department is implemented as an app with models, views, URLs, templates, and reports. Shared workflow components connect patient timeline, service orders, payment events, notifications, and audit records.

Layer	Responsibility
Presentation	Django templates, Bootstrap/Jazzmin, print templates, responsive layout
Application	Role decorators, form validation, workflow views, report exports
Domain	Patients, visits, doctors, referrals, billing, departments, staff, payroll
Data	SQLite development / PostgreSQL production using environment settings
Operations	Backups, deployment docs, Redis/Celery-ready configuration

3. Installation Guide

Local installation is supported through run.bat and run.sh. These scripts create a virtual environment, install dependencies, copy the environment template, migrate the database, create demo accounts, and start the server.

- Install Python 3.11 or 3.12.
- Run run.bat on Windows or run.sh on Linux/macOS.
- Use demo password: password.
- Open /accounts/login/ for staff and /portal/login/ for patients.

4. Deployment Guide

Production deployment should use `DEBUG=False`, PostgreSQL, a reverse proxy such as Nginx/Apache, HTTPS, static collection, media storage, background workers, and scheduled backups.

- Set `SECRET_KEY` and `ALLOWED_HOSTS`.
- Configure PostgreSQL variables in `.env`.
- Run `migrate` and `collectstatic`.
- Use Gunicorn/Uvicorn/Daphne as appropriate.
- Use Redis for Channels/Celery workloads.

5. PostgreSQL Setup

Switching to PostgreSQL requires no code changes. Set `DJANGO_DB_ENGINE=postgres` and provide `POSTGRES_DB`, `POSTGRES_USER`, `POSTGRES_PASSWORD`, `POSTGRES_HOST`, and `POSTGRES_PORT`.

- Create production database and user.
- Run migrations on PostgreSQL.
- Dump/load development data only through documented procedures.
- Configure backups before go-live.

6. User Roles and Permissions

Role	Access Summary
Main Super Admin	Full system, settings, reports, backup, staff, finance, clinical/admin modules
Finance	Revenue, payroll, salary, attendance reports, finance exports
Department Head	Own-department staff, attendance, leave, referrals, scoped reports
Registration/EHS	Patient lookup, registration, OPD/EHS visits
Doctor	Consultation, patient search, referrals, own profile
Department Users	Own module dashboard and operational reports
Staff	Own profile, attendance, leave, notifications
Patient	Portal profile, bookings, bills, visits, reports

7. Patient Registration and OPD

Patient Registration and OPD is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Patient Registration and OPD dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

Integration Points

- Patient identity is shared across modules to avoid duplicate entry.
- Billing and reports consume operational records from department workflows.
- Notifications and referrals move work between doctors, departments, finance, and counters.
- PostgreSQL-ready migrations support production deployment.

Verification Notes

- Smoke-tested pages returned HTTP 200 during final review.
- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

8. Patient Portal

Patient Portal is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Patient Portal dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
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9. EHS / Extension Services

EHS / Extension Services is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- EHS / Extension Services dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
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Verification Notes

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- Relevant PDF and Excel reports were verified where applicable.
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10. Doctor Consultation

Doctor Consultation is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Doctor Consultation dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
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Verification Notes

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- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

11. Referrals and Service Orders

Referrals and Service Orders is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Referrals and Service Orders dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

Integration Points

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Verification Notes

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- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

12. Billing and Cash Counter

Billing and Cash Counter is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Billing and Cash Counter dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

Integration Points

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Verification Notes

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- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

13. Finance and Accounts

Finance and Accounts is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Finance and Accounts dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

Integration Points

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Verification Notes

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- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

14. Laboratory

Laboratory is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Laboratory dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

Integration Points

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- Notifications and referrals move work between doctors, departments, finance, and counters.
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Verification Notes

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- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

15. Radiology

Radiology is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Radiology dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

Integration Points

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Verification Notes

- Smoke-tested pages returned HTTP 200 during final review.
- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

16. Pharmacy

Pharmacy is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Pharmacy dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

Integration Points

- Patient identity is shared across modules to avoid duplicate entry.
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Verification Notes

- Smoke-tested pages returned HTTP 200 during final review.
- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

17. Admissions and Ward

Admissions and Ward is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Admissions and Ward dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

Integration Points

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Verification Notes

- Smoke-tested pages returned HTTP 200 during final review.
- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

18. Nursing

Nursing is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Nursing dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

Integration Points

- Patient identity is shared across modules to avoid duplicate entry.
- Billing and reports consume operational records from department workflows.
- Notifications and referrals move work between doctors, departments, finance, and counters.
- PostgreSQL-ready migrations support production deployment.

Verification Notes

- Smoke-tested pages returned HTTP 200 during final review.
- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

19. Operation Theatre

Operation Theatre is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Operation Theatre dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

Integration Points

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- Notifications and referrals move work between doctors, departments, finance, and counters.
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Verification Notes

- Smoke-tested pages returned HTTP 200 during final review.
- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

20. Blood Bank

Blood Bank is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Blood Bank dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

Integration Points

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- Billing and reports consume operational records from department workflows.
- Notifications and referrals move work between doctors, departments, finance, and counters.
- PostgreSQL-ready migrations support production deployment.

Verification Notes

- Smoke-tested pages returned HTTP 200 during final review.
- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

21. Insurance

Insurance is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Insurance dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

Integration Points

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Verification Notes

- Smoke-tested pages returned HTTP 200 during final review.
- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

22. Medical Records

Medical Records is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Medical Records dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

Integration Points

- Patient identity is shared across modules to avoid duplicate entry.
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Verification Notes

- Smoke-tested pages returned HTTP 200 during final review.
- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

23. Staff Attendance

Staff Attendance is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Staff Attendance dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

Integration Points

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Verification Notes

- Smoke-tested pages returned HTTP 200 during final review.
- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

24. Leave Management

Leave Management is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Leave Management dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
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Integration Points

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Verification Notes

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- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

25. Payroll and Salary

Payroll and Salary is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Payroll and Salary dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
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Verification Notes

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- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

26. Reports and Exports

Reports and Exports is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Reports and Exports dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

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Verification Notes

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- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

27. Notifications

Notifications is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Notifications dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
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Verification Notes

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- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

28. Audit Trail

Audit Trail is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Audit Trail dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
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Integration Points

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Verification Notes

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- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

29. Backup and Restore

Backup and Restore is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Backup and Restore dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

Integration Points

- Patient identity is shared across modules to avoid duplicate entry.
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- Notifications and referrals move work between doctors, departments, finance, and counters.
- PostgreSQL-ready migrations support production deployment.

Verification Notes

- Smoke-tested pages returned HTTP 200 during final review.
- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

30. PostgreSQL Deployment

PostgreSQL Deployment is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- PostgreSQL Deployment dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

Integration Points

- Patient identity is shared across modules to avoid duplicate entry.
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Verification Notes

- Smoke-tested pages returned HTTP 200 during final review.
- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

31. Security and Maintenance

Security and Maintenance is implemented as part of the integrated hospital workflow. It uses Django models, role-restricted views, templates, forms, and database migrations. Data is linked to the central patient/staff identity records where applicable.

Operational Scope

- Security and Maintenance dashboard/page access is controlled by role permissions.
- Search, filtering, printing, and reporting are connected where applicable.
- Records are preserved for audit and reporting instead of being deleted from operational history.
- Barcode-driven lookup is used for patient/staff/department documents where implemented.

Integration Points

- Patient identity is shared across modules to avoid duplicate entry.
- Billing and reports consume operational records from department workflows.
- Notifications and referrals move work between doctors, departments, finance, and counters.
- PostgreSQL-ready migrations support production deployment.

Verification Notes

- Smoke-tested pages returned HTTP 200 during final review.
- Relevant PDF and Excel reports were verified where applicable.
- Permissions were reviewed for Super Admin, Finance, Department Head, staff, and department users.

32. Database Design

The database is Django ORM-driven and database-agnostic. Key entities include User, Role, Patient, Visit, Doctor, Department, Referral, ServiceOrder, Bill, PaymentEvent, StaffAttendance, StaffLeaveRequest, StaffSalaryPayment, and departmental records.

- One Patient = One Hospital ID.
- Visits are chronological under the same patient.
- Staff and patient accounts are separate.
- Billing records are preserved for audit.

33. ER Diagram Explanation

The ER diagram PDF included with the deliverable summarizes relationships between patients, visits, doctors, departments, referrals, bills, payments, staff, attendance, leave, payroll, and department-specific records.

34. DFD and Flowcharts

The DFD explains movement of information from registration to departments, billing, finance, reports, and medical records. Flowcharts describe OPD, referral, billing, attendance, leave, payroll, and backup processes.

35. API Documentation

- Patient barcode lookup: /patients/api/lookup/?code=PATIENT_CODE
- Staff barcode lookup: /accounts/staff/api/lookup/?code=STAFF_ID
- Attendance device punch: /accounts/staff/attendance/device-punch/?staff_id=STF000001&direction=in
- Radiology template endpoint: module-specific JSON template endpoint where implemented.

36. Troubleshooting Guide

- If Django is missing, activate the virtual environment and install requirements.
- If migrations are pending, run `python manage.py migrate`.
- If PostgreSQL fails, verify DB credentials and network access.
- If static files are missing in production, run `collectstatic`.
- If payment gateway fails, verify live/UAT merchant credentials.

37. Maintenance Guide

- Run backups daily.
- Review audit logs.
- Update dependencies in a staging environment first.
- Run automated checks before deployment.
- Monitor PostgreSQL, disk, CPU, RAM, and application logs.

38. Future Enhancements

- Real fingerprint SDK integration after hardware selection.
- Live payment gateway certification.
- Advanced BI dashboards.
- SMS/OTP provider integration.
- Cloud object storage for media/backups.
- High availability database replication for large hospitals.

39. Implementation Appendix 1

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

40. Implementation Appendix 2

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

41. Implementation Appendix 3

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

42. Implementation Appendix 4

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

43. Implementation Appendix 5

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

44. Implementation Appendix 6

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

45. Implementation Appendix 7

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

46. Implementation Appendix 8

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

47. Implementation Appendix 9

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

48. Implementation Appendix 10

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

49. Implementation Appendix 11

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

50. Implementation Appendix 12

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

51. Implementation Appendix 13

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

52. Implementation Appendix 14

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

53. Implementation Appendix 15

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

54. Implementation Appendix 16

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

55. Implementation Appendix 17

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

56. Implementation Appendix 18

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

57. Implementation Appendix 19

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

58. Implementation Appendix 20

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

59. Implementation Appendix 21

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

60. Implementation Appendix 22

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

61. Implementation Appendix 23

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

62. Implementation Appendix 24

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

63. Implementation Appendix 25

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

64. Implementation Appendix 26

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

65. Implementation Appendix 27

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

66. Implementation Appendix 28

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

67. Implementation Appendix 29

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.

68. Implementation Appendix 30

This appendix records production implementation notes, verification expectations, user training points, and maintenance guidance for hospital deployment teams.

- Confirm role access before go-live.
- Train users on barcode scanning and patient identity reuse.
- Verify printer alignment at each counter.
- Run backup and restore test in staging.
- Review reports with Finance and Super Admin.
- Document department-specific SOP decisions.